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PERSPECTIVES ON…

Using Games to Make Formative Assessment Fun in the Academic Library

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Assessment is everywhere in higher education. It is no longer acceptable to assume students are learning what is taught in the classroom or the library; administrations and accrediting bodies want to see evidence of learning. In libraries, this often happens in the form of major evaluation projects such as LibQual+®, or locally-created equivalents. More specifically for library instruction, there are standardized assessments such as Project Sails, iSkills, and ILT. These all have an important role to play in assessing student satisfaction and learning, but they are all summative macro-assessments and quite intimidating to librarians and students alike. They often require significant resources in the form of time and money and do not allow for immediate improvements of student learning. More mini-assessments are needed in the library classroom while there is still time to improve instruction. For assistance in this endeavor, librarians can learn from the literature on formative assessment and game-based learning.

This article makes two arguments. The first is for increased awareness of formative assessment among instruction librarians. The second is that educational games have the potential to make good environments for formative assessment. There are solid bodies of literature available on each subject, though little on formative assessment in the library literature. In analyzing the literature on the pedagogical qualities of each, remarkable similarities emerged. This article analyzes these similarities in detail and uses two information literacy games to demonstrate how formative assessment can be embedded in educational library games.

INTRODUCTION TO FORMATIVE ASSESSMENT

In contrast to summative assessments, which happen after the learning is complete and when students can no longer improve their performance, formative assessments are "in-the-process-of-learning assessments (Baker & Delacruz, 2008, p. 23)." These types of assessments are a "continuous flow of accurate information on student learning (Angelo & Cross, 1993, p. 3)," and happen while learning is still taking place so that instructors can adjust their instruction to best meet the current learning needs of their students. They are informal, discrete, and frequent, as often as every few minutes. Examples of formative assessments include discussion, interpretation of nonverbal cues, pre-tests, summary writing, think–pair–share, and using clickers or hand-signals for class polls. In each of these cases, it is only formative assessment if the instructor uses the feedback to modify instruction (Black & Wiliam, 1998). Formative assessment can also be considered a dynamic conversation between instructors and learners. Just as formative assessment provides the teacher with feedback, students are also given feedback, allowing them to receive an answer to the question "how am I doing?" rather than just "how did I do (Rolfe & McPherson, 1995)?"

Formative assessment is widely accepted as an effective practice and it is heavily emphasized among education practitioners in the primary and secondary levels. Black and Wiliam (1998) conducted an extensive literature review of several hundred articles on formative assessment. They found a large number of quantitative studies where cultivation of formative assessment practices led to significant learning improvement. They concluded that the effect of this policy change was larger than most other educational interventions.

While all educators can benefit from learning about formative assessment, it is perhaps particularly useful to instruction librarians as we usually are restricted to "one shot" sessions with students. This means that our time with students is particularly limited for the content we have to share with them, we usually do not have time to develop relationships with the students, and we do not often see the final research projects. Each of these factors hinders our ability to get feedback on how much of the content they have absorbed. Consciously working formative assessment into our limited time allows students and librarians to exchange meaningful feedback while engaging both at the same time.

Because formative assessment is so prevalent in the education literature and because it has the potential to work well within the confines of the typical library classroom, it is surprising that there is so little information in the library literature about it. The author was able to identify only two articles with titles containing "formative assessment" in library journals. Dunaway and Orblych (2011) used a pretest and mid-instruction assessment in order to adjust their instruction with MBA students. Seely, Fry, and Ruppel (2011) closed an instruction session with pre-service teachers by asking students to list one thing they learned and one thing they still had questions about. The librarians followed up with the students on these reported

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questions outside of the scheduled bibliographic instruction session. This could be considered lower-level "formative assessment" as it was used to correct confusion after the library session rather than while learning was still taking place. In this case, students had to be deprogrammed for any misconceptions they adopted (Angelo & Cross, 1993). Ideally, formative assessment corrects learning within the library session, although this remains difficult when limited to an hour or two with students.

FORMATIVE ASSESSMENT IN EDUCATIONAL GAMES

The 2012 Horizon Report (Johnson, Adams, & Cummins, 2012) lists educational games as one of the up-and-coming technologies that will transform education. Educational games provide a unique opportunity to embed formative assessment into the library classroom in a fun way. By combining educational games with pre- and post-tests, multiple studies show it to be at least as effective as other methods of assessment, and students prefer them over comparative methods (Coleman, Livingston, Fennessey, Edwards, & Kidder, 1973; Delacruz, Chung, & Baker, 2010). A prominent video game scholar argues that "A [commercial] video game is just an assessment. All you do is get assessed, every moment, as you try to solve a problem. And if you don't solve it, the game says you fail, try again. And then you solve it, and then you have a boss, which is a test, and you pass the test (Ellis, 2008)." The formative assessment is what is going on throughout most of the game as students test out new skills, get feedback, adjust strategies, and advance to new levels. The "boss fight" is the summative assessment, the "one that counts."

A number of articles and prominent scholars in the education and game studies literature analyze how games can be used for assessment. Delacruz et al. (2010) looked at a mathematics game called *Puppetman* to teach units and addition to summer school students. Game performance was significantly related to pre- and post-test scores and the students reported wanting to continue playing the game even outside of the classroom. Schiller (2008) demonstrates how the commercial game *Portal* uses a technique called "gating" (also known as "choke points") to ensure players do not progress beyond their abilities by chance.

Two articles specifically focus on games for formative assessment. Baker and Delacruz (2008) emphasize the importance of designing games and assessment activities to support learning goals. Assessments can focus on the process, rather than just reaching the goal. Therefore, recording the time it takes to complete each task and number of attempts at each task, not just the end result, are examples of formative assessment. Hudson and Bristow (2006) analyze the use of *Who Wants to Be a Millionaire* in a medical class. After each question was answered by the student in the "hot seat," the entire class showed what answer they had each chosen and a tutor led a discussion. This game demonstrated two common formative assessment activities, response cards and discussion (Fisher & Frey, 2007), and the authors highlight the need to maintain a safe environment for discussion.

INSTITUTIONAL BACKGROUND AND GAMES

Lycoming College is a small, private liberal arts college of approximately 1400 students. The Snowden Library has a strongly established games program that is well documented in the scholarly literature. This article will focus on two that are directly related to information literacy. The small size of our campus is important to the context in which these games were designed and implemented. They were each designed by one or two librarians with a very small budget that included a book on Flash programming, small prizes for raffles to encourage survey participation, and minimal printing costs.

Secret Agents in the Library was the first structured game, which was designed as a general introduction to library research in four sections of freshman composition (Broussard, 2010). This was designed to be played in a one-hour class in the library and mixes online activities and resources with a physical exploration of the library. In the game, students work in small groups of rookie agents to catch an intruder who has broken into the "information mainframe" (a. k. a. the library). In the process of catching the spy, they explore reference materials, books, databases, and the MLA citation style. The game is followed by a debriefing activity using clickers. This game was successfully used for several semesters until the professor of a majority of these sections left the college and his successor used a more specific assignment that was not compatible.

In the fall of 2009, we launched an online plagiarism game called *Goblin Threat* (Broussard & Oberlin, 2011). This game was designed to create a positive environment in which students can learn about plagiarism and how to avoid it. In the game, the campus has been taken over by plagiarism goblins. The player must save the campus by clearing each room of its hidden goblins, each goblin asking one question about plagiarism. This entirely-online game was marketed to our faculty as something that could be played as homework, with a printable certificate as proof of completion. The game has been used hundreds of times among our own students. Furthermore, as the material is not institution-specific, over seventy other high school media centers, college libraries, and English departments are linking to it from their own websites. Statistics for page visits are only available from January 2011 to the present, but in that span of time the page has been visited over 54,000 times.

EIGHT SHARED ELEMENTS OF FORMATIVE ASSESSMENT AND GAME-BASED LEARNING

The following section addresses eight remarkable similarities in the literature between formative assessment and game-based learning, and illustrates how these were demonstrated in the three information literacy games at Lycoming College. Most of these elements are in themselves widely considered to be effective qualities in the classroom. Each of these eight qualities interacts closely with each other in complex ways, as illustrated in Fig. 1. The author of this article argues that because there is such an overlap in the major elements of these two methods of instruction, one could conclude that games make good environments for formative assessment, thus allowing assessment to be fun for the instructor and the students.

ACTIVE LEARNING

Formative assessment and game-based learning each have an inherent assumption of being incorporated into an active learning environment. Active learning, in its many forms, is so integral to both, that all of the other elements that will be discussed fall within its domain, as shown in Fig. 1. Formative assessment only works with active learning (Black & Wiliam, 1998; Shepard, 2005) as formative assessment is a dynamic conversation between teacher and students. It requires students to be testing their understanding, providing feedback to the instructor, and interpreting feedback received from the instructor. In lecture-based learning and most online tutorials, there is no conversation. The information flow is one way, from the teacher or tutorial to the student, until the time of the quiz at which point neither the instruction nor the learning can be modified for improvement. Requiring students to be active participants throughout the learning process has been shown to improve student performance, engagement, and motivation (Cauley & Mcmillan, 2009).

It is important to note that while active learning and formative assessment are so closely intertwined that it is often hard to distinguish between the two, that they are not the same thing. Active learning can and does happen without formative assessment. This happens when the instructor is not looking for evidence of student understanding and providing feedback. Many articles on active learning will admit that the

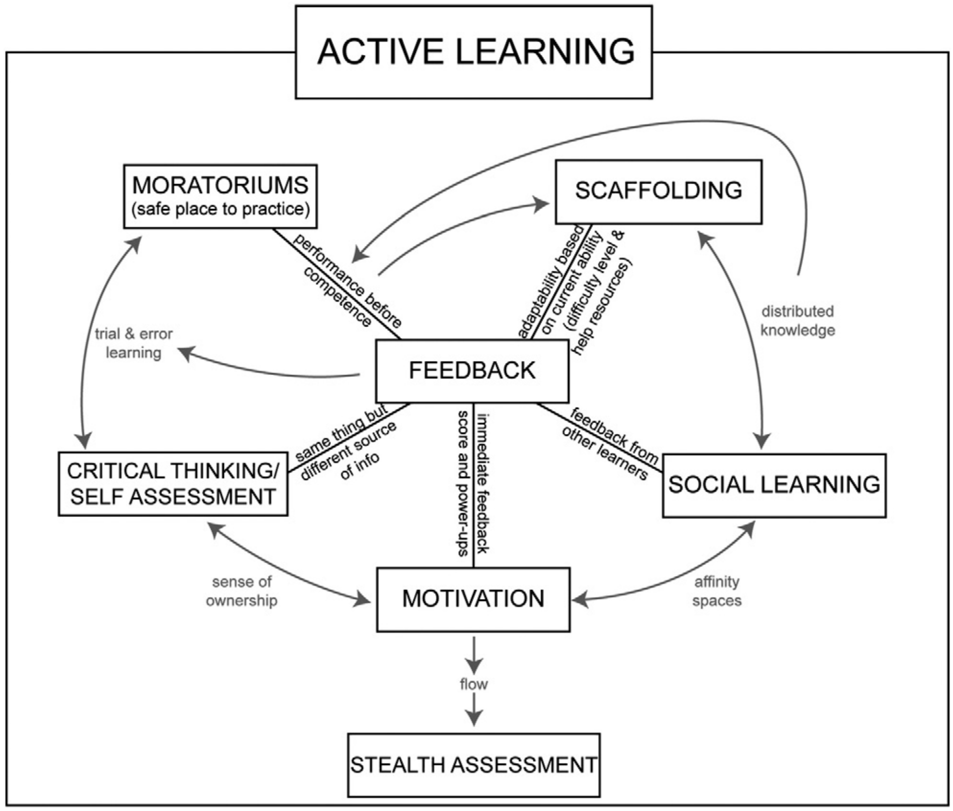

Fig. 1. Interactivity between elements shared by game-based learning and formative assessment.

activities must be *meaningful* (Cooperstein & Kocevar-Weidinger, 2004), and one can easily make the argument that active learning is done best in conjunction with formative assessment.

Just as formative assessment must take place in an active learning environment, games are inherently interactive and also involve an adaptive conversation between the player and the game world. Games require the player to *participate*. This may take the form of rolling dice and moving a piece on a board, answering trivia questions, or pressing buttons on an electronic device. Furthermore, nearly all games require players to be mentally active in assessing options, risks, and making decisions.

Furthermore, games and formative assessment each involve *practicing* skills learned and *experiencing* the world in new ways. Sadler writes, "few physical, intellectual or social skills can be acquired satisfactorily simply through being told about them. Most require practice in a supportive environment (Sadler, 1989, p. 120)." Experiential learning and problem-based learning are forms of active learning that focus on experience. This emphasis on experience often requires the student to act before he or she has all of the required skills. Gee (2008) calls this "performance before competence," in relation to video games. Similar skills are described by Sadler (1989) in regards to developing students' ability to self-moderate through formative assessment. In both games and formative assessment, the gap between the skills the students/players have and the ones they need to perform at the goal level is compensated for through help materials or through dispersed knowledge (to be discussed later) until they develop those skills within themselves.

Formative assessment calls for additional activeness on the part of the instructor as well. The instructor is continuously looking for feedback from students and acting on that feedback. Formative assessment encourages a "conscious presumption against pre-determined content of […] instruction (Dunaway & Orblych, 2011, p. 35)," meaning that instructors cannot plan a class assuming they understand what the students are starting with or will absorb. It turns the classroom from a one-way conversation where the instructor is the leader, to a two-way dialog where the instructor is a facilitator and participant, albeit one with the advantage of experience. It also requires the instructor to be flexible enough to adapt to the unique personality of each class (Angelo & Cross, 1993). Instructors must be constantly "on their toes" in a formative assessment classroom and willing to adapt to student feedback. For educational games design, the game must demonstrate this activeness in place of the instructor.

Both of the Snowden Library's information literacy games required the player to be an active participant by reading, processing, and acting on their understanding of the material. In each of these cases, the game is the central instructional activity, preceded by nothing but instructions and followed by nothing but review. *Secret Agents in the Library* required groups to explore electronic resources, answer questions, and eventually find physical books and print journals. It was designed to use many of the help materials that are always available to library users, such as the library directory and maps. Moreover, the game links to many online resources such as the library's website and online databases, requiring students to get hands-on practice with the resources they will later need for their assignment. *Goblin Threat* did not require students to be literally active as it is entirely online. However, students must be mentally active to find the hidden goblins, read questions, and decide which answers are the best. Because both games were well-structured online games, the "activeness" of the instructor had to be programmed into the game. This was done by anticipating players' moves in a beta version, then testing the games on colleagues and student workers for suggestions on refinement.

FEEDBACK

Feedback is a central ingredient of formative assessment and game-based learning alike. It "informs learners of their present state of learning, so they can plan action to close the gap between their present state of learning and their desired goal(s) (Hudson & Bristow, 2006, p. 33)." Examples of feedback in the classroom include the teacher's written and verbal comments on student work, and students' non-verbal cues, discussion, written work, and interactions with each other. Without feedback, teachers would not know what the learners have understood and students would not have clues to improve their performance. Most learning in online games comes from trial-and-error learning, which cannot take place without feedback (Schiller, 2008). Feedback in games comes in many forms such as points, power-ups, virtual clothes and tools, access to new game spaces, leaderboards, and help materials. Ideally, feedback in both formative assessment and games should be specific, frequent, immediate, and adaptive.

The need for specific feedback is most often mentioned in the formative assessment literature. Kluger and Denisi (1996) found that feedback interventions that were vague, such as lavish, non-specific praise, often hurt student performance. Sadler (1989) and Black and Wiliam (1998) say that feedback should focus on a) the goal b) the student's current position in relation to the goal and c) how to close the gap between the two.

Many game-based learning and formative assessment scholars stress the need for frequent and immediate feedback (Gee, 2003; Kickmeier-Rust & Albert, 2010; Squire, 2011). Fisher and Frey (2007) say that formative assessment should happen no less than every fifteen minutes. Young (2010), who was writing about what professors can learn from video games, equates frequent feedback to subgrades on a paper so that students know what they are doing right and what they need to improve. Discussion of immediate feedback is often tied to what Millennial students have come to expect, but is also closely tied with pedagogy and motivation. Immediate or timely feedback "reduces the sense of distance between the player's efforts and successes (Annetta, Minogue, Holmes, & Cheng, 2009, p. 76)." Students/players associate cause and effect more easily if the two are chronologically closely associated. While immediate feedback is not always possible in games or in the classroom, short turnaround times should always be considered a top priority.

Feedback needs to be adaptive based on students' ability (Kickmeier-Rust & Albert, 2010). Sadler (1989) argues that because formative feedback is so qualitative, it cannot be automated by a computer. However, since he wrote his article over twenty years ago, commercial video games have partially proved him wrong. While the refinement of some skills may remain elusive to computerized feedback, video games expertly react to the players' ability by adjusting difficulty level and customized help materials as are appropriate based on players' input. One example of adaptive feedback in educational games is a miniature tutorial after a wrong answer is chosen by the player. The player can use this feedback in subsequent attempts to correctly complete that game activity. In the classroom and non-digital games, this is done by a skillful and flexible teacher or game facilitator. For example, when an instructor realizes during class discussion that many of the students have misinterpreted a key concept, he or she can clarify that misconception or ask one of the students to clarify it for the rest of the class.

Formative assessment requires teachers to inspire critical thinking by moving away from factual questions with one correct answer, to questions that have many possible answers. The teacher must quickly adapt to students' unconventional answers, bringing out what is correct about them, and using those questions to further the class activity. Help materials and coaching are generous when skills are immature, and scale back as the student/player develops. Feedback from the teacher or game slowly becomes self-assessment as students/players takes responsibility for their own learning and performance goals and have the experience to judge themselves (Sadler, 1989).

The online element of each information literacy game included feedback in the form of mini-tutorials after incorrect answers. In *Secret Agents in the Library*, correct answers were rewarded with a harp sound-effect and an increase in score. Incorrect answers led to a gong sound effect, and often explained why the answer was incorrect in one sentence. During the exercise where students had to locate a particular print book, the librarian was available in the stacks to provide assistance in reading call numbers for any groups that were struggling. At the end of the game, while the intruder always ended up behind bars (after all, players have several activities where they *must* provide the correct answer in order to proceed), the final page gave a message that varied depending on the team's final score. Teams that scored highly were notified that they had been promoted from rookies to full agents, while teams that did poorly were told that being an agent was not right for them and invited them to work with the librarian for a better understanding of the library. To this date, all teams have obtained the promotion.

In *Goblin Threat*, players are also given feedback on correct and incorrect answers. Correct answers lead to a splat sound effect and the word "Correct!" on a cartoonish green blood splat image, then the slayed goblin disappears. Incorrect answers get a sound effect of Homer Simpson saying "d'oh!", a mini tutorial leading them to the correct answer, and an invitation to try again. The goblin does not disappear, and the student can attempt each question as many times as is necessary to slay each goblin. When each room is completely clear of goblins, a door opens and the student is invited to progress to the next room. *Goblin Threat* also offers a progress report in the form of a notification of how many goblins remain in the current room and a book icon titled "Rooms Cleared." When players hover over the book, they can see how many rooms they have cleared and how many remain. This gives students a sense of achievement and takes away the mystery of how much remains to be accomplished.

MOTIVATION

Oka (2005) defines motivation as the "force behind behavior (p. 330)." Motivation is required to learn, and motivation to learn is often divided into two varieties. Extrinsic motivation is driven by external rewards such as candy, prizes, or grades. Intrinsic motivation takes place without external rewards, and behavior is done for its own sake. Intrinsic motivation to learn is preferred because learners demonstrate more satisfaction and enjoyment in learning and willingly take on more challenging tasks, thus reaching higher academic achievement. Intrinsic motivation plays a significant role in formative assessment and game-based learning.

There can be little doubt that those who engage in commercial video games have strong intrinsic motivation because players clearly spend a lot of time, money, and energy to play when they are not required to (indeed, even when they are discouraged from playing). Gee's (2003) seminal book *What Video Games Have to Teach Us About Learning and Literacy* asserts that players work hard to learn, and enjoy even very difficult mental challenges in commercial video games. The motivation comes from a feeling of self-agency and satisfaction of accomplishments. These players are so motivated that their desire to engage in the game goes beyond the game itself to "affinity spaces" (often online) where they share ideas and even create new media such as in-game modifications and fan fiction.

Cauley and McMillan (2009) state that formative assessment "is now recognized as one of the most powerful ways to enhance motivation (p. 1)." This is primarily because students come to take responsibility for their own learning, which is intrinsically motivating (Cauley & Mcmillan, 2009; Rolfe & McPherson, 1995; Sadler, 1989; Shepard, 2005). As the primary element of formative assessment is instruction and learning modification, the students share control of the class with the instructor. Furthermore, as self-assessment skills develop through the formative assessment process, students gain a sense of self-agency, which is important for encouraging intrinsic motivation (Cauley & Mcmillan, 2009; Oka, 2005).

It is important to note that the relationship between extrinsic and intrinsic motivation is not as clear-cut as their definitions suggest in games or formative assessment. External rewards are not always apparent, or the reward may be significantly delayed. Intrinsic motivation often involves satisfaction, which could be considered an abstract external reward. Furthermore, a learner may simultaneously have intrinsic and extrinsic motivation. This clarification is important to this discussion for multiple reasons. Games offer multiple types of rewards. Points and competition can be considered extrinsic rewards. Yet as points have no real-world value, and competition can remain friendly, they can also provide intrinsic motivation. In the classroom, the need for grades on tests and formal writing is very real, yet many students still enjoy the process of learning as well. Educational games tied to a class are most effective when required as part of a participation grade, or offered for extra credit points (Broussard, 2012). Students who strongly wish to avoid the game should be given an alternate assignment if possible, as game-based learning research shows that games are not for all learners (Edery & Mollick, 2009). Competition can increase motivation but only if efforts are made to ensure a safe environment of *friendly* competition between players where the winner may earn a relatively small extra reward, but others are not penalized.

Measuring motivation is difficult, particularly for games that are required activities. Evidence of engagement and motivation is often shown through non-verbal cues among players. In *Secret Agents in the Library*, students were observed holding their hands like pistols and singing the James Bond theme song. They were role-playing more than the game required them to. They were clearly engaged as final scores were posted for each group. Whether or not a score is involved, players will strive for the bragging rights of having the highest score. Friendly competition is a powerful motivator.

As *Goblin Threat* is usually played outside of the classroom, players cannot be observed. However, a small focus group of students were involved in the development of this game. They clearly expressed enjoyment with the game, even those who were tired of the content. Furthermore, a satisfaction survey was included at the end of the game when it was first released to the general campus population. A number of participants described the game as fun or preferring this method of education to those they had previously experienced.

MORATORIUMS

Not all assessments in the classroom need to be graded. The goals of learning, and even the assessment of learning, often contradict the goals of grading (Sadler, 1989). Gee (2003) introduces the term “psychosocial moratoriums” to describe safe areas in video games where players can practice their new skills with reduced consequences before entering the game where those skills count. In the formative assessment literature, there is an emphasis on ungraded assessments, non-judgmental and stress-free learning spaces (Hudson & Bristow, 2006; Rolfe & McPherson, 1995). This allows students to practice, experiment, and take risks without the fear of humiliation or hurting their grades. This comes in many forms, such as ensuring a safe environment for discussion (Hudson & Bristow, 2006) and assigning “low-stakes” writing activities (Fisher & Frey, 2007).

Video games and formative assessment encourage students to start acting on new skills before they are truly capable of succeeding. With practice and guidance, students will become competent. But a safe environment to practice is critical. Gee (2003) condemns traditional education for not providing safe places to practice new skills. In lecture-based learning, students are told what they should know, and then tested on it. Students must be encouraged to put a lot of effort into learning, yet this is difficult when they must commonly divide their focus between learning and grades. They cannot be expected to experiment and take risks if there is a good probability of being unsuccessful the first time. Failure and the critical thinking required to overcome it can lead to the best learning, but is impossible if those initial failures appear on report cards. This problem can be solved through formative assessment, where students have activities designed to practice, take risks, and transfer knowledge from one situation to another, then receive feedback on the results, before moving to summative assessments.

Many librarians do not have the option of giving grades, particularly for the individual one-hour, assignment-specific classes we teach. This removes the temptation to grade formative assessment activities. However, librarians still must offer chances for students to practice new skills even if they are imperfect, and they must facilitate a non-threatening environment where participants feel respected. One way library games in their entirety can be designed as moratoriums is to allow the player to make as many attempts as is necessary to succeed. While this relieves some of the urgency that makes games so appealing, it does make them better teaching tools.

Neither of the library games were graded, although one freshman composition professor offered one or two extra credit points to team members with the highest score for *Secret Agents in the Library*. The few extra credit points were enough to inspire friendly competition, but not enough to make this a high-stakes activity. Neither of these games was timed or required a great deal of hand–eye coordination. *Goblin Threat* offered an unlimited amount of attempts at each question. This, combined with the mini-tutorials after incorrect answers, nearly guarantees eventual success. However, they are designed so that reading and thinking about each question carefully allows players to complete the game successfully in less time than randomly clicking buttons. During the introduction and review activities outside of each game, an effort was made by the librarian to maintain a safe learning environment where all honest contributions were valued.

SCAFFOLDING

Shepard (2005) argues that there is no difference between formative assessment and scaffolding. In each, students incorporate new knowledge with their previous knowledge and experiences, gradually developing increasingly complex skills. Commercial video games adeptly demonstrate scaffolding. Players often start in a moratorium where they can safely practice how to move, then each level requires them to add to their skillset a little bit at a time, all of which leads up to the major “boss fight” at the end.

Angelo and Cross (1993) write about how goals can be used in formative assessment. While goals are often thought of as terminal points, they can also serve as reference points. While there may be one big goal in a unit or in a game, there are many smaller and very specific goals that must be met along the way. They use the analogy of navigating the ocean, where the end goal is the port, but along the way the navigator will use the stars as reference points to make sure the ship stays on track. If there is indication they are not on track, they will remediate to get them back before the ship becomes completely lost. These mini goals that accumulate and lead toward the final goal are another way of interpreting the idea of scaffolding.

Adaptive feedback from the teacher or the game allows the players to always be challenged without becoming bored because the teacher or game is always using evidence of the players' current abilities before offering help or moving on. Teachers who are good at formative assessment can avoid setting goals that are too high or too close to students' current abilities. “If the learner perceives the gap is too large, the goal may be regarded as unattainable […] Conversely, if the gap is perceived as too small, closing it might be considered not worth any additional effort (Sadler, 1989, p. 130).” Gee (2003) and other video game scholars (Kirriemuir & McFarlane, 2012; Smith & Baker, 2011) have similarly commented on good games' ability to keep players on the edge of mastery, moving the bar just a little bit higher as each mini goal is met so that they are constantly challenged but never bored.

Many video and online games demonstrate scaffolding by increasing difficulty levels as skills increase. *Goblin Threat's* questions were not designed to get harder as the game progressed, although the last level is designed to be a review. There are mini-goals, though, in that students are focused on clearing out one room at a time before gaining access to the next play area. *Secret Agents in the Library* demonstrates scaffolding through simulations of how research builds on itself as the player/researcher progresses from reference books, to books, to articles, to citation. For example, there is a Boolean logic exercise where players see six spies wearing colorful hats and trench coats. They are asked to click on each spy that would fit the search "blue AND green," then the process is repeated for "blue OR green." The next activity within the game asks players similar questions that apply Boolean logic more literally in the databases for a hypothetical research topic.

CRITICAL THINKING AND SELF-ASSESSMENT

Reflection, combined with experience, is a critical part of the educational process (Black & Wiliam, 1998; Nicholson, 2012). Formative assessment and game-based learning both emphasize reflection and critical thinking. Whether for problem-based learning or the latest popular video game, critical thinking is mandatory in problem solving. It is also mandatory for self-assessment, another important part of formative assessment and games.

The literature on formative assessment stresses the need for educators to rethink what kinds of questions they ask their students. Too often we simply ask factual questions with one right answer. Students struggle to guess what their educators are thinking and hesitant to respond only to find out they were wrong. Furthermore, many instructors do not give students enough time to think about the answer before providing the answer for them. Both of these discourage critical thinking in the classroom. Formative assessment pushes teachers to ask open-ended questions and allow for unorthodox answers (Black & Wiliam, 1998). Similarly, good games offer and accept multiple solutions to each problem (Gee, 2003; Shute, 2011), challenging students and sparking creativity and critical thinking.

In video games, players have a clear goal from the beginning, they want to win. They are constantly assessing points, health levels, amount of ammunition, and many other sources of information. While this is feedback in the sense that the information comes from the game, the game does not often tell the player what to do with this information. In that sense, interpreting the feedback in games can be considered self-assessment. In some video games, the game starts by interpreting information for the player, for example giving them a reminder to seek more ammunition, but discontinues the practice as the player gets used to doing this on his or her own.

In formative assessment, self-assessment does not come as naturally. Teachers must guide students into accepting responsibility for their own learning goals (Sadler, 1989). Students cannot be expected to do this on their own at first because they lack the skills and experience to do so. While video game players adopt their goals in the beginning and students' self-adoption of learning goals must be developed, the similarities continue in that careful coaching can lead them from frequent feedback to well-developed self-assessment skills. Sadler (1989) explains that feedback and self-assessment are identical except that feedback comes from the instructor (external to the student) whereas self-assessment comes from within the students themselves.

Games that simply test the players' knowledge are not effective at inspiring critical thinking. This is a frequent criticism of "edutainment" or "gamification" in the classroom. Indeed, this may be the area where library games struggle the most as they are in their infancy and even the most effective games are very simple (Broussard, 2012). For games involving technology, librarians are often limited by their own self-taught coding skills to multiple-choice, matching, and short-answer questions. For real-world games, it is often a question of being stuck in the mindset of factual question–answer mode or not knowing how to design game-like activities that involve critical thinking. Yet even with the technological limitation, a change in mindset and skillful question development can inspire reflection.

Additionally, educational games are most effective when combined with wrap-around assessments, meaning formative assessment activities that are outside of the game itself. Many game facilitators know this best as "debriefing" (Nicholson, 2012). This means that many limitations in the game design can be compensated for through reflective activities such as discussion, journaling, or role playing as a game designer. In any debriefing, the facilitator (instructor) should focus on players' initial feelings, analysis of the game, how it relates to previous knowledge, and what to do with the information learned (Nicholson, 2012). The main purpose of educational games is for players to be able to transfer what they learned in a game to real-life experiences. Coleman et al. (1973) found that stronger students were more capable of doing this on their own than weaker students, and that debriefing helped the weaker students transfer their new knowledge to practical applications.

Incorporating critical thinking into the online portion of these two games was a challenge with limited programming skills. However, carefully crafted questions can still require students to think critically about fixed-answer questions. A library science student intern with several years' experience as a high school English teacher collaborated with the full-time librarian to develop thought-provoking multiple-choice, true/false, and sorting questions for *Goblin Threat*. The sorting questions had multiple correct answers, and required the player to evaluate each possible answer.

Secret Agents in the Library also asked questions with multiple correct answers, such as ones that required students to identify all of the subject-specific resources among six or seven titles. Furthermore, because it required players to explore real resources and online tools, it compelled them to think critically about these tools. For example, they are directed to the library website and told to find a database that will help them identify relevant reference book articles. With some critical thinking, they should identify that either the *Reference Portal* or the *Databases* links will lead towards the correct answer (*Reference Universe*).

SOCIAL LEARNING

Social learning is not a part of every game or formative assessment activity, but is a frequent trait that deserves attention. From when video games first entered pop-culture as arcade games to the rise of social media and console-networking technology, video games have always been social. Players can compete in real-time from anywhere in the world, and the Internet is the perfect place for "affinity spaces (Gee, 2005)," communities dedicated to specific games where players can post their knowledge and even share original fan fiction, artwork, or game modifications. Many public libraries and even some academic libraries hold video game tournaments as outreach events because players love to come together to play.

Just as game players often learn as much from their fellow players as from the game itself, social learning in the classroom allows students to learn from each other and not just directly from the instructor. Many of the activities mentioned in Fisher and Frey's book (2007) include social learning. Examples of social learning in formative assessment include:

- Discussion
- Think–pair–share: students are given a moment to think about their own response, share it with one other person, then share it with the class
- Clickers, hand signals, or response cards
- Interactive writing: students take turns with the teacher writing on the board
- Peer-review of writing.

Sadler (1989) states that peer-review is the best tool to develop students' self-assessment skills because the feedback they get from each other is more like what they strive to give themselves. He also points out that many of these activities have the secondary benefit of granting students rich and immediate feedback while at the same time relieving the instructor of the time-consuming responsibility of writing comments on all student work.

Social learning also allows for what Gee (2003) calls "dispersed knowledge (p. 176)," meaning that the player or student does not need to have all of the knowledge and skills themselves. Instead, players/students work with the game system and together, each contributing their own skills and knowledge into a central place. In many educational games, this can be accomplished simply by having players work in small teams to complete the game activity. In *Secret Agents in the Library* (Broussard, 2010), teams were observed discussing answers and sharing knowledge. Final scores for teams were much higher than scores of any individual student would have been because team members were able to pool their knowledge and discuss the results.

Social learning can also be accomplished through debriefing. Allowing the class to discuss a game after the gameplay allows learners to share their observations, feelings, and any confusion. Through such discussion, students can learn from each other and gain more from the game than their own limited experience allowed. *Secret Agents in the Library* was followed by a debriefing activity using clickers. Students were asked a series of review questions. If nearly the entire class got the question correct, they moved on to the next question. If there was any evidence of confusion from the clicker data, the librarian gave a mini-lecture. This shows how clickers can be used for formative assessment as part of the critical debriefing or review activity.

It is worth noting here that another library game is built around social learning, and that is Michigan University's *Bibliobouts* (Markey et al., 2010). This game simulates the research-gathering process by requiring players to submit citations for a class paper, then judge their own and each other's for quality and appropriateness. While there is a fair amount of set-up required by the instructor (usually a professor rather than a librarian), nearly all in-game assessment is shared between the game system and other players. The game system provides scores based on quantitative measures such as how many citations or tags were entered. Other players provide the qualitative judgments that are important to formative assessment based on the quality of each other's submissions.

STEALTH ASSESSMENT

Shute uses the term "stealth assessment" to describe evaluation that has been embedded into educational games so that it does not interrupt the players' game experience. It is "seamlessly woven directly into the fabric of the instructional environment to support learning of important content and key competencies (Shute, 2011, p. 504)." Engaging video games make players forget how hard they are working to play, which is why such games are so popular and commercially successful.

Stealth assessment is closely related to most of the previously mentioned qualities and requires feedback in order to happen. However, it is through motivation that stealth assessment is possible because it is the intrinsic motivation and "flow" of an activity that hides the fact that assessment is taking place. "Flow" describes the state where a person is so absorbed in an activity that he or she forgets everything else, including a sense of time and self-awareness (Csikszentmihalyi, 2004). Flow happens when high challenges are combined with high levels of skill. Many educational game scholars (Kirriemuir & McFarlane, 2012; Shute, 2011; Squire, 2011) have tied this idea of flow to commercial video games, but it can happen in the classroom as well.

It is precisely because the term "assessment" is so dreaded by students and teachers alike that stealth assessment has so much appeal. In formative assessment, the goal is that the instructor is fully aware of the assessment taking place while students are so engaged in learning that they are unaware of it. When class activities and assignments are not being graded, the feedback does not seem like "evaluation," rather it seems more like "help." Through social learning and opportunities to demonstrate critical thinking, students are not aware that their instructor is using their responses as feedback to gauge their understanding. Because assessment has been seamlessly integrated into the classroom, it fades into the background for the student, and perhaps allows more accurate assessment to happen.

Using games for formative assessment furthers the ability to hide assessment and make it fun for the instructor and the students. Educational game designers strive to create assessments that closely relate to the desired learning objectives without breaking a sense of flow (Shute, 2011). However, putting this into action is quite complicated and even controversial. Knowledge of what makes a good educational game is still in its infancy and few educators have the technical knowledge, time, creativity, or money to equal those of the commercial video games our students play for leisure. However, in a previous study of digital library games (Broussard, 2012), the author found that in most cases it was not the most expensive or technologically impressive games that led to the most satisfaction. Instead, it was the games that were best-suited for the intended purpose. Many librarians know enough about pedagogy and are creative enough to develop fun classroom games even if their technical and artistic abilities are limited. This is echoed in the literature by several educational game designers who found students to be very forgiving of games' primitive appearance, but equally unforgiving for bad design (Annetta et al., 2009; Smith & Baker, 2011).

Players were not as aware that they are being assessed in *Goblin Threat* or *Secret Agents in the Library* as they would have been with a test. These games were designed to mimic familiar casual games the players were already familiar with, such as the escape-the-basement genre of online games or *Where in the World is Carmen Sandiego?* When students are enjoying the game experience, as many clearly were based on observation and survey data, they are more focused on the *play* and less on the *work*. Humor was also used to add motivation and mask the assessment, such as the silly places where goblins are hidden in *Goblin Threat* or the self-destructing messages in *Secret Agents in the Library*. Some educators may have difficulty seeing these as assessments, just as those same educators may already be using formative assessment in their classrooms and having difficulty considering it assessment.

CONCLUSION

There is a distinct difference between using games to assess students, as discussed here, and assessing the games themselves. This article has focused on how games can assess students formatively. Each of these games were assessed in various ways. *Goblin Threat* was assessed using a small focus group in its testing phase, then releasing it to faculty for feedback, then providing a link to a satisfaction survey during the first few weeks after its public debut. While some did not enjoy the game or thought the goblins were too hard to find, overall players were pleased with the new avenue for learning plagiarism content. As previously mentioned, *Secret Agents in the Library* was assessed through a review activity that provided insight into students' knowledge of library research after playing the game. Students did well on the multiple-choice clicker questions and many reported having fun.

In order to use games for formative assessment, serious thought must be given to the games' assessment methods very early in the design process, preceded only by the selection of learning objectives for the game. Shelton and Scoresby (2011) suggest creating a storyboard for the game (a common practice for story-based games), listing the learning objectives, and then physically connecting the learning objectives to the corresponding activity on the story board. Coleman et al. (1973) stress the need to make learning objectives central to succeeding in the game.

Games are not appropriate for all learning objectives. Games are time-consuming to develop and less forgiving of design flaws than other methods of instruction. Small mistakes in game design can easily make the game break down. For librarians teaching one-shot library sessions, getting buy-in from the course professor is critical. If the professor is not interested in game-based learning, it is not for the librarian to override his or her opinion. However, in such a case, this article still provides useful information on formative assessment, which can be incorporated into any classroom situation.

Based on the numerous benefits of formative assessment described in the education literature, the paucity of similar articles in the library literature is strong evidence that instruction librarians may be missing a fundamental solution to improving student learning in the library classroom. While formative assessment is happening in many library classrooms, it needs to be increased and librarians need to be more conscious of doing it systematically.

Both formative assessment and game-based learning have many advantages for the library one-shot class, particularly in regards to increasing intrinsic motivation. As it is normally difficult to encourage intrinsic motivation in such a short window of time, these two instructional methods provide much-needed suggestions that librarians can quite easily adopt in the library classroom. There is evidence in the literature and demonstrated in the analysis of the two information literacy games here that formative assessment is already happening in games and that students are demonstrating learning and engagement. Increasing our knowledge of these two teaching methodologies can lead to rewarding and enjoyable learning experiences, both for the students and the librarian.

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